

The number of good polynomials is $\boxed{35}$.

Proof:

A polynomial $P(x)$ is good if it satisfies three conditions:

1. It is monic of degree 7 with integer coefficients.
2. The absolute values of all coefficients are odd and strictly less than 2025^2 .
3. The absolute values of all coefficients are roots of P .

Step 1: Analyzing the roots and coefficients.

- Since $P(x)$ is monic, the leading coefficient is 1. Thus, $|1| = 1$ must be a root, so $P(1) = 0$.
- Let the coefficients of $P(x)$ be $1, c_6, c_5, \dots, c_0$. The absolute values $|c_i|$ for $0 \leq i \leq 6$ must be roots of $P(x)$.
- All roots of $P(x)$ are integers (since their absolute values are integers, and non-integer roots would have non-integer absolute values not among the coefficients' absolute values).

Step 2: Sum of coefficients condition.

- From $P(1) = 0$, we have $1 + c_6 + c_5 + \dots + c_0 = 0$.
- Let $S = c_6 + c_5 + \dots + c_0$. Then $S = -1$.

Step 3: Coefficients must be ± 1 .

- Suppose any coefficient c_i has $|c_i| \geq 3$. Then $|c_i|$ must be a root, but constructing such a polynomial while satisfying $S = -1$ and the root conditions leads to contradictions (e.g., requiring additional roots not covered by coefficients' absolute values). Thus, all coefficients must have $|c_i| = 1$, i.e., $c_i \in \{1, -1\}$.
- Therefore, each c_i is either 1 or -1 , and their sum $S = -1$.

Step 4: Counting valid coefficient configurations.

- Let k be the number of $+1$ coefficients among $c_6, \dots, c_0$. Then the sum of these coefficients is $k - (7 - k) = 2k - 7$.
- Setting $2k - 7 = -1$ gives $k = 3$. Thus, exactly three of the seven non-leading coefficients must be $+1$, and the remaining four must be -1 .
- The number of ways to choose these three positions is $\binom{7}{3} = 35$.

Step 5: Verifying all conditions.

- **Monic and degree 7:** Satisfied by construction.
- **Absolute values are odd and $< 2025^2$:** All coefficients are ± 1 , so their absolute values are 1, which is odd and less than 2025^2 .
- **Absolute values are roots:** The absolute value 1 is a root (as $P(1) = 0$). Other roots (e.g., -1) do not affect this condition since only the coefficients' absolute values need to be roots.

Conclusion: There are $\boxed{35}$ good polynomials.